Aerospace Science and Technology 122 (2022) 107327

Contents lists available at ScienceDirect

Aerospace Science and Technology

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Cross-flow transition model predictions of hypersonic transition research vehicle

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ARTICLE INFO

Article history:
Received 21 April 2021
Received in revised form 8 December 2021
Accepted 1 January 2022
Available online 7 January 2022
Communicated by Qiulin Qu

Keywords:
Hypersonic flow
Lifting body
Cross-flow transition
Transition modeling

ABSTRACT

Hypersonic transition research vehicle (HyTRV) is a customized lifting body for studying hypersonic three-dimensional boundary-layer transition on a complex geometry. In this study, the transition model of an HyTRV lifting body at relatively high angles of attacks is investigated. A hypersonic cross-flow transition correlation is embedded in the γ-$\widetilde{\mathrm{Re}}_{\theta t}$ transition model on the Chant2.0 computing platform, called the C-γ-Re_θ transition model. The transition model is validated in a series of typical three-dimensional aerodynamic configurations. HyTRV's flow characteristics and transition phenomenon are investigated. The model predictions are in good agreement with the stability analysis results and shockwave wind tunnel flow visualizations, exhibiting the model potential for hypersonic cross-flow transition prediction. The results of wind tunnel experiments, stability analysis, and model predictions confirm that transition regions on the upper/lower surface of HyTRV are dominated by cross-flow instabilities. With an increase in the angle of attack, the circumferential pressure gradient on the lower surface decreases and cross-flow intensity reduces; the transition onset location recesses until finally disappears. 2–3 pairs of isolated cross-flow transition zones exist on the surface of the HyTRV at a low angle of attack, which guarantees the HyTRV to be a typical validation case for cross-flow transition modeling.

1. Introduction

The accurate prediction of hypersonic three-dimensional (3D) boundary layer transition induced by cross-flow instability is a challenging issue in the aerodynamic design of hypersonic vehicles. In real flight, the boundary layer often has a velocity component perpendicular to the direction of the inviscid streamline called cross-flow because of the influence of the 3D configuration of the aircraft and attitude angle [1]. The cross-flow induced instability is the cause of 3D boundary layer transition on large parts of the surface of hypersonic vehicles [2].

Most engineering applicable transition prediction techniques are RANS-based [3]. Further, the RANS-based turbulence model is the main transition prediction method in this work. In the field of cross-flow transition modeling, researchers built several low-speed models according to different cross-flow distinguishing method [2]. The cross-flow extensions and original γ-Re_θ models are mainly designed for subsonic swept flat plates and swept wings [4][5], and can not be directly used for hypersonic cross-flow predictions [6]. Zhang et al. [7][8] provided some hypersonic correlations on the SST-γ-Re_θ model. In the field of hypersonic cross-flow transition model, a considerable amount of work is conducted around the k–ω–γ model [9][10][11][12], the capability of cross-flow transition prediction is acquired by introducing non-turbulent fluctuation eddy viscosity and cross-flow timescale. Zhang et al. [13] improved the γ-Re_θ transition model for the transition prediction on the sharp and elliptic cones with $AoA = 0°$, which is the basic work for this cross-flow extension. Xu et al. [14][15] proposed a single-equation transition/turbulence model to predict the transition on a prolate spheroid and sharp cones. Xu et al. [16] proposed a fully localized two-equation transition model considering laminar stability theory. However, the γ–Re_θ–related models still do

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https://doi.org/10.1016/j.ast.2022.107327
1270-9638/

Nomenclature

γ	intermittency factor
$\widetilde{Re}_{\theta t}$	transition onset momentum thickness Reynolds number
AoA	angle of attack
p	pressure
p_0	stagnation pressure
p_∞	freestream pressure
$Re_{crossflow}$	cross-flow Reynolds number
$\delta_{10\%}$	wall-normal direction distance at 10% Wmax
δ	boundary layer thickness
U	local velocity
Ma	Mach number
Ma_∞	freestream Mach number
Re_∞	freestream Reynolds
T_0	stagnation temperature
T_∞	freestream temperature
T_w	wall temperature
Tu	turbulence intensity
Tu_∞	freestream turbulence intensity
L/D	lift-to-drag ratio
$P_{\theta t}$	production terms of $\widetilde{Re}_{\theta t}$ transport equation
D_{CF}	cross-flow destruction terms of transport equation
$H_{crossflow}$	non-dimensional cross-flow strength
$\vec{\Omega}$	vorticity vector
$\Omega_{streamwise}$	streamwise component of vorticity vector
h	surface roughness
C_1, C_2	closure coefficients in transition model
N	N value of stability theory
N_{crit}	critical N value at transition location
St	Stanton number
u_{cf}	cross-flow velocity
u_s	stream-wise velocity
Re_{CF}	cross-flow transition criterion
μ	molecular viscosity
μ_t	eddy viscosity
W_{max}	max cross-flow velocity

Abbreviations

DNS	Direct Numerical Simulation
FSC	Falkner-Skan and Cooke equations
FSTI	Free Stream Turbulence Intensity
HyTRV	Hypersonic Transition Research Vehicle
LES	Large Eddy Simulation
LST	Laminar Stability Theory
RANS	Reynolds-averaged Navier-Stokes
TOL	Transition Onset Location
TSP	Temperature Sensitive Paint

not have the ability of hypersonic cross-flow prediction although the γ–Re_θ model is widely used in the industry.

Considering the broad application prospects of γ–Re_θ in the engineering field and the strong demand of hypersonic cross-flow prediction, a hypersonic cross-flow extension on γ–Re_θ is presented. The hypersonic transition model in this work is based on the subsonic transition model framework proposed by Langtry [17]. The experimental database of the transition criterion is extended by the stability theory. The fully localized transition criterion is built considering the cross-flow intensity and the effect of surface roughness. A hypersonic cross-flow extension on the modified γ–Re_θ transition model is implemented on the Flow Solver Chant2.0, which is named the C–γ–Re_θ transition model [6][18], where C represents the Chant platform and cross-flow. After being validated through the cases of a sharp cone and an elliptic cone, the investigation of transition on the HyTRV lifting body with C–γ–Re_θ model is conducted.

2. Cross-flow transition on hypersonic vehicles

Fig. 1 illustrates the main-flow and cross-flow profiles in a 3D boundary layer flow. Under the effect of the surface curvature and circumferential pressure gradient, the streamline outside the boundary layer begins to curve and the centrifugal force of the fluid particle on the potential streamline in the direction of the vertical streamline is balanced with the pressure gradient. The unbalance of centrifugal force and pressure gradient inside the boundary layer results in a secondary flow perpendicular to the direction of the potential flow streamline. This flow component is called cross-flow. Fig. 1 illustrates the cross-flow velocity and mainstream velocity distributions along the wall-normal direction, where $u_s(y)$ denotes the main-flow profile that is consistent with the direction of the streamline outside the boundary layer, and $u_{cf}(y)$ denotes the cross-flow profile perpendicular to the outside streamwise direction. The cross-flow velocity profile has a generalized velocity inflection point; the cross-flow velocity profile will lead to nonviscous instability (inflection point instability) and the

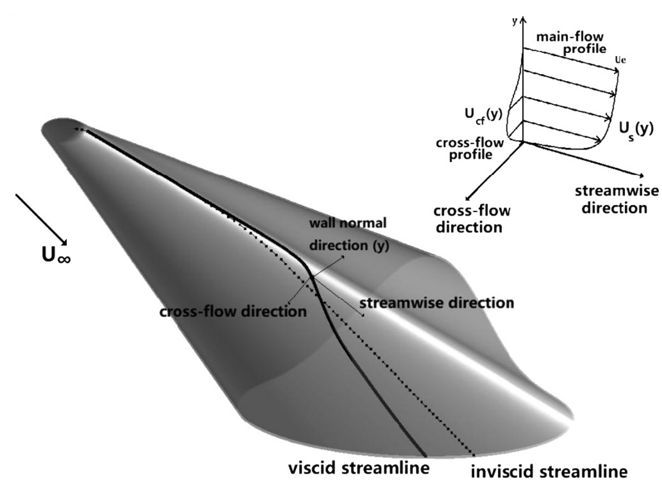

Fig. 1. Schematic of cross-flow profile on HyTRV lifting body.

large growth rate of this instability form will lead to an earlier transition.

Cross-flow induced transition can be divided into the following important stages according to current research: receptivity, linear and non-linear growth of disturbance, secondary instabilities, and breakdown. In low-speed swept flat-plate experiments, Bippes [19] found that the transition process is dominated by the stationary cross-flow vortex at the low-inflow turbulence intensity and dominated by the unsteady cross-flow vortex at the high-inflow turbulence intensity. In hypersonic flow, there is a clear gap of the HIFiRE-5 transition onset shape between the quiet inflow and noise inflow conditions [20]. If the transition model is designed to predict aircraft, the transition data refer the quiet tunnel and flight test. Unfortunately, the related validation cases of the sharp cone and the HyTRV vehicle are not quiet inflow conditions at present due to experimental limitations.

3. C-γ-Re$_\theta$ transition model

The γ–Re_θ transition model is the most widely used fully localized transition model in engineering applications. The model was applied to low-speed transition flows such as wind turbines, aircraft wings, and turbomachinery blades. The wide application and excellent low-speed performance of the model are reasons why we select it as a basic transition model. Several extended studies have been conducted based on this model [6][13][18][21]. The streamwise vortex intensity can be approximately used to reflect the cross-flow intensity. For example, the Helicity parameter [22] is an indicator of cross-flow intensity. Langtry et al. [17] extended the γ–Re_θ transition model with a stationary cross-flow transition criterion considering the surface roughness effect on the cross-flow transition.

We need to guarantee that the basic transition model predicts hypersonic streamwise transition correctly and the cross-flow extension is developed for hypersonic conditions for achieving the model predictions of hypersonic 3D transitions by cross-flow extensions. Whether the original γ–Re_θ model or the cross-flow extension was developed for low-speed transition predictions, its empirical correlations are validated through low-speed cases. The model's incapability of hypersonic streamwise transition predictions and that of the hypersonic cross-flow predictions of the above cross-flow extension has been previously ascertained [6][21].

In this work, a hypersonic cross-flow transition model is extended based on a hypersonic modified γ–Re_θ transition model on Chant2.0 [13]. The cross-flow transition criterion [6] is designed based on the transition flow field predicted by DNS [23] and LST. The model construction and validation cases are discussed in this part.

3.1. Model construction

The hypersonic cross-flow extension is constructed on an enhanced γ–Re_θ model [13] by inserting a destruction term into the $\widetilde{\mathrm{Re}}_{\theta t}$ transport equation. The cross-flow transition criteria are developed based on hypersonic cross-flow transition data [23]. The details of the extension are as follows:

The basic transition model is the γ–Re_θ [17] transition model with two transport equations. The γ-intermittency equation and local transitional onset momentum-thickness Reynolds number $\widetilde{\mathrm{Re}}_{\theta t}$ equation are the two transport equations. Consistent with the form of Langtry's cross-flow extension [17], the hypersonic cross-flow extension on γ–Re_θ transition model is implemented by adding a destruction term D_{CF} to the transport equation of $\widetilde{\mathrm{Re}}_{\theta t}$. The transport equations are formulated as

$$\frac{\partial(\rho\gamma)}{\partial t}+\frac{\partial(\rho U_j\gamma)}{\partial x_j}=P_\gamma+E_\gamma+\frac{1}{\mathrm{Re}_\infty}\frac{\partial}{\partial x_j}[(\mu+\frac{\mu_t}{\sigma_f})\frac{\partial\gamma}{\partial x_j}]$$

$$\frac{\partial(\rho\widetilde{\mathrm{Re}}_{\theta t})}{\partial t}+\frac{\partial(\rho U_j\widetilde{\mathrm{Re}}_{\theta t})}{\partial x_j}=P_{\theta t}+D_{CF}+\frac{1}{\mathrm{Re}_\infty}\frac{\partial}{\partial x_j}[\sigma_{\theta t}(\mu+\mu_t)\frac{\partial\widetilde{\mathrm{Re}}_{\theta t}}{\partial x_j}]$$

The destruction term D_{CF} is defined as:

$$D_{CF}=c_{CF}\cdot\rho\min(\mathrm{Re}_{CF}-\widetilde{\mathrm{Re}}_{\theta t},0)\cdot F_{\theta t}$$

The other source terms are described in detail in [17].

The γ–Re_θ transition model has already been improved for hypersonic streamwise-flow transition phenomena [13], i.e., pressure-gradient parameter and turbulent Prandtl number corrections. Hypersonic improvements ensure model prediction for transition on the cones at the 0° angle of attack, as shown in Fig. 2; however, the model continues to lack the ability for cross-flow transition prediction at an angle of attack [13]. Therefore, the hypersonic cross-flow transition is implemented.

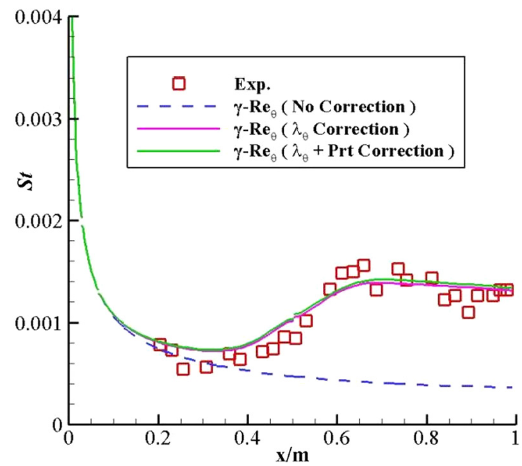

Fig. 2. Heat transfer distribution for sharp straight cone ($\mathrm{Ma}_\infty=8$) [13]. (For interpretation of the colors in the figure(s), the reader is referred to the web version of this article.)

Quiet tunnel experiment results [20][24] demonstrate that the hypersonic cross-flow transition location can be greatly influenced by surface roughness height. This factor should be considered when the model is constructed. Because of the complexity of the transition mechanism, it is not realistic to build a transition model purely based on the transition mechanism. Thus, we still select to make an empirical correlation based on a large amount of experimental data. One difficulty in transition modeling is the lack of experiment data; i.e., the experimental results of the quiet tunnel and flight test wherein surface roughness changes only. The surface roughness height is the root mean-squared height of the surface acquired by measuring instruments. The definition of surface roughness height is

$$h=\sqrt{\frac{1}{n}\sum_{i=1}^{n}(h_i-\overline{h})}$$

where n, hi, and $\overline{h}$ denotes the number of the measured points, single measurement, and mean of measured heights, respectively. The roughness height of a smooth surface is of the order of micrometers.

In this study, the DNS data of transition under different surface roughness [23] are extended with the e–N method of the stability theory [18]. In other words, the DNS data is used for N_{crit} calibration, and the e-N method of LST is adopted to predict the transition data of Fig. 3. The empirical transition criterion is calibrated based on the above dataset of "critical cross-flow Reynolds number (Re_{CF}) and surface roughness (h)" (Fig. 3). Re_{CF} is defined as the cross-flow Reynolds number at the cross-flow transition onset location. It is worth noting that the empirical calibration line can be shifted up and down due to local cross-flow intensity. Although the influence mechanism of surface roughness on cross-flow transition is not clear yet, the surface roughness effect can be introduced into the model in the form of a correlated logarithmic relationship with the critical criterion Re_{CF}. Namely, with inflow conditions unchanged, the decrease in surface roughness will lead to an increase in Re_{CF} and a retreat of cross-flow *TOL* (Transition Onset Location). The hypersonic transition criterion with surface roughness effect is set as follows, which is similar to the form of the low-speed one [17]. The value of Re_{CF} is considerably smaller than that of the

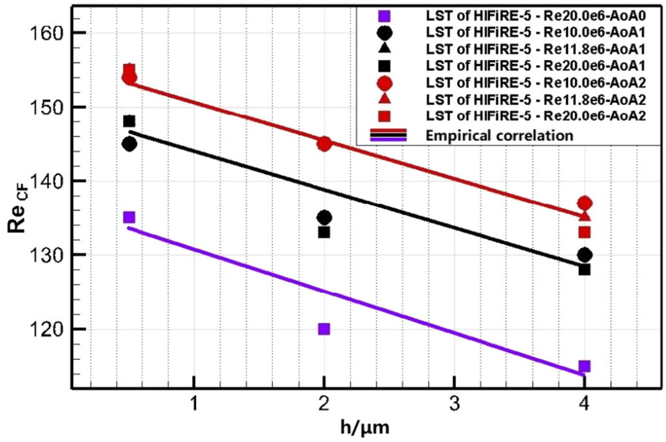

Fig. 3. LST predicted transition data of the elliptic cone [18].

Table 1
Cross-flow extension coefficients & parameters.

C_1	C_2	c_{CF}	l_μ
-9.618	128.33	0.2	1 µm

low-speed [17] which means the low-speed cross-flow model may be invalid in the hypersonic conditions. Similar to low-speed flow, the data shows a relatively strong linear relationship in the logarithmic coordinate system. We use the data at a 0° angle of attack as the benchmark data; the cross-flow criterion is formulated as

$$\mathrm{Re}_{CF} = C_1 \cdot Ln(\frac{h}{l_\mu}) + C_2 + f(H_{crossflow})$$
$$f(H_{crossflow}) = 6000\left|0.1066 - \Delta H_{crossflow}\right| + 50000(0.1066 - \Delta H_{crossflow})^2$$
$$\Delta H_{crossflow} = H_{crossflow}(1.0 + \min[1.0\frac{\mu_t}{\mu}, 0.4])$$

Re_{CF} is not the commonly defined cross-flow Reynolds number; the common cross-flow Reynolds number is a nonlocal quantity. Transition is predicted by comparing the cross-flow Reynolds number with its critical value. The criterion Re_{CF} is a localized approximation of the critical $Re_{crossflow}$ and is compared with $\widetilde{\mathrm{Re}}_{\theta t}$ obtained from the transport equation. The $H_{crossflow}$ denotes the nondimensional cross-flow strength [17] defined as follows where U denotes the local velocity and $\Omega_{streamwise} = \left|\vec{U} \cdot \vec{\Omega}\right|$. The definition of the shift-up-function $f(H_{crossflow})$ is the same as that reported in ref. [18]. Cross-flow extension coefficients and parameters are summarized in Table 1. The C_1 and C_2 are obtained by the least square method with transition data [18], c_{CF} is validated to be 0.2 in NLF(2)-0415 swept wing cases [6][25]. Alongside the cross-flow parameters, the other parameters can be found in [13][17]. The model is mainly used for cases of sharp cone, elliptic cone and, lifting body with Ma $= 6$, $Re \in [1.0 \times 10^7, 4.2 \times 10^7]$ and $AoA \in [0°, 8°]$. In all the cases, the angle of yaw is zero.

3.2. Validation cases

The selection of numerical schemes for hypersonic conditions refers to the relevant research of Chen's work [26]. An AUSMPW+ scheme with minmod limiter and second-order monotone upstream-centered schemes for conservation laws (MUSCL) scheme are utilized. Second-order central difference schemes are implemented for viscous flux discretization. The lower-upper symmetric Gauss-Seidel (LU-SGS) scheme approach is used for implicit time marching. All schemes are employed on the in-house Flow Solver-Chant2.0. The reliability of Chant2.0 for laminar, turbulence, and transition flow simulations have been validated and verified by a series of numerical simulation cases [13][21][25]. Validation cases for the hypersonic cross-flow transition model C–γ–Re_θ include wind tunnel experiments of the 7° half-angle sharp cone [27] and the quiet wind tunnel experiments of HIFiRE-5 [20]. The predicted transition onset location (*TOL*) with the hypersonic cross-flow effect is compared with those experimental results to ensure the hypersonic cross-flow transition predicting the capability of the extended model.

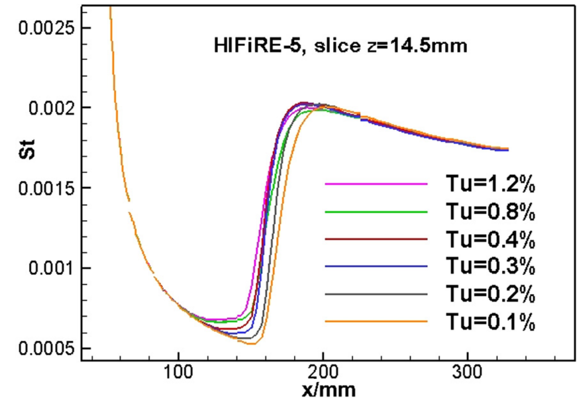

Fig. 4. *St* distributions of different *FSTI* for HIFiRE-5.

In transition modeling, the freestream turbulence intensity (*FSTI*) is used to quantify the inflow disturbance level although in hypersonic wind tunnel the noise level (NL) is usually measured instead. The *FSTI* is estimated from the measured noise level (NL) of wind tunnel. The sound induced turbulence intensity is calculated with the formula: $Tu_{noise} = \frac{NL}{\gamma Ma}$. This essential correlation between sound induced turbulence intensity (Tu_{noise}) and pressure fluctuations (NL) was derived from typical disturbance wave's analysis in Euler's system for hypersonic conditions [32]. The total turbulence intensity Tu_{total} is empirically estimated to be twice as much as Tu_{noise}, *FSTI* $= Tu_{total} = 2 \times Tu_{noise}$, since the main source of hypersonic inflow disturbance comes from acoustic noise. It has to be noted that the measured noise level (NL) is only used for *FSTI* estimation. In transition modeling, the *FSTI* is used to quantify the inflow disturbance level. The measured noise level can not be equivalent to turbulence intensity but a certain level of fluctuation only.

The model's sensitivity of *FSTI* is analyzed. For the condition of quiet tunnel and general tunnel, the *FSTI* is set from 0.1% to 1.2%. Fig. 4 shows the sensitivity of HIFiRE-5's *TOL* to *FSTI* in transition modeling ($\mathrm{Ma}_\infty = 6$, $\mathrm{Re}_\infty = 9.5\mathrm{e}6$), namely a similar *Staton* number distributions under different *FSTI*. The relative error of *TOL* is less than 4% when *FSTI* $\leq 0.4\%$. When $0.4\% \leq$ *FSTI* $\leq 1.2\%$, the relative error of *TOL* reaches 7% when compared with low disturbance income conditions.

3.2.1. Wind tunnel experiments of a sharp cone

Infrared thermography experiments of the 7° half-angle sharp cone were conducted in the $\phi = 1$ m hypersonic wind tunnel at CARDC [27]. The freestream conditions are illustrated in Table 2 and other parameters can be inquired in [27]. The inflow turbulence intensity is not measured but estimated from the noise level for transition modeling. Grids on the solid wall and symmetry plane of the sharp cone are shown in Fig. 5. Half model grids are selected and contain about 0.67 million hexahedral cells. The resolution level in the normal direction of the wall is optimized to limit y+ no greater than 1.0.

Fig. 6 compares the near-wall and potential-flow streamlines of the sharp cone at angles of 2°. Owing to the existence of cross-

Table 2
Freestream conditions of the sharp cone.

Ma_∞	Re_∞ (/m)	*Noise Level*	Tu_∞ (for modeling)	h_{rsm} (μm)	T_∞ (K)	T_w (K)	p_∞ (Pa)	α (◦)
6	1.0×10^7	3.93%	0.8%	0.2	57.81	300	696.7	0 2 4 6

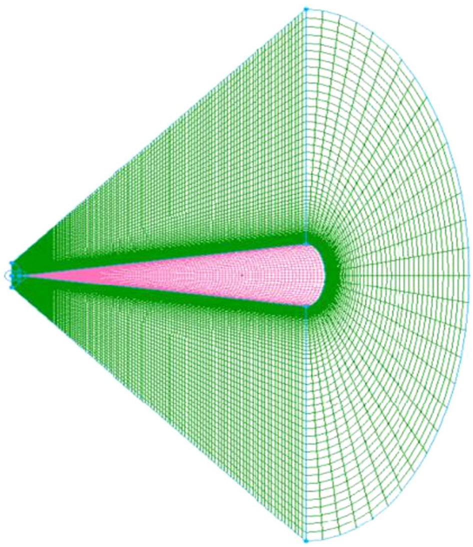

Fig. 5. Computed mesh for the sharp cone.

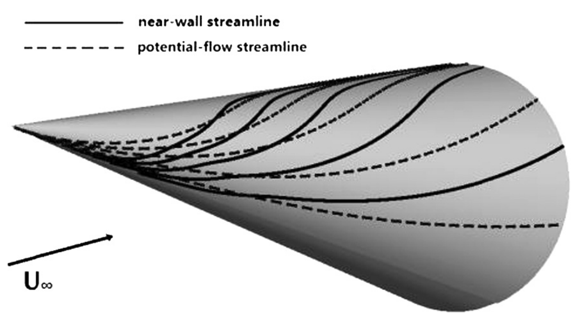

Fig. 6. Schematic of near-wall streamline and potential-flow streamline on the sharp cone.

flow velocity, two types of streamlines intersect on the side of the sharp cone between leeward and windward. Further, cross-flow induced transition exists in this area. The computational Stanton number compared with experimental temperature distributions in Fig. 7 indicates that the cross-flow transition onset location on the sharp cone is relatively accurately predicted by $C-\gamma-Re_\theta$ model at different angles of attack.

Table 3 summarizes the model predicted transition location, experimental transition location and their discrepancy in percentage terms. The *TOL* of experiments is provided from [27], and the model predicted *TOL* is determined by *St* distributions at the intersection point of the transition curve's tangent and the extension of laminar curve. The deviation of original model from experiments of Cases II-IV is invalid for their overall transition trends are opposite from experiments.

3.2.2. Quiet wind tunnel experiments of an elliptic cone

The testing models of hypersonic cross-flow transition experiments are simple geometries such as sharp cones and elliptic cones. HIFiRE-5 has a geometry where the hypersonic flow exhibits cross-flow transition at 0° angle of attack with a relatively strong spanwise pressure gradient. The dimension parameters have been reported elsewhere [20]. The freestream conditions are illustrated in Table 4. The inflow turbulence intensity is estimated from the noise level. The surface roughness range is from 0.17 μm to 0.42 μm and 0.3 μm is used in the transition model; other parameters can be found [20]. The mesh on the solid wall and symmetry plane of the elliptic cone are shown in Fig. 8. Further, half model grids are selected to economize the computational costs. Half model contains about 2.5 million of cells and near-wall cell size in the wall normal direction are strictly limited.

A comparison of the quiet tunnel and model prediction results are shown in Fig. 9. The transition onset location of the transition model matches the experimental results well at various *Re*. Although the striped transition details cannot be predicted, the overall shape fits well. There is a discrepancy on leading edge between experiments and numerical results. It is mainly due to the low observed experimental heat flux caused by the deviation of the surface's normal direction from the observed direction.

4. Crossflow transition prediction of HyTRV lifting body

4.1. HyTRV lifting body

The total length and width of HyTRV in wind tunnel are 800 mm and 300 mm, the swept angle is 9.34 degrees. The windward surface's cross section is designed to be an elliptical curve with the ratio of the long to the short axis to be 4:1, and the leeward surface is designed as a brimmed hat shape with the CST (class function and shape function transformation) technique. The shape of HyTRV is promulgated to public through authorized use (e-mail address for authorization application: ghtu@skla.cardc.cn).

The HyTRV vehicle has more transition mechanisms than sharp cones and elliptic cones for its more complex three-dimensional geometry. Stability analysis indicates [29] that there are several relatively independent cross-flow zones and streamwise vortex structures. The HyTRV cases can be used to evaluate the cross-flow prediction model.

The stability analysis of the HyTRV [29] is compared with model predictions in this chapter. As shown in Fig. 10, the upper surface and lower surface can be divided into several zones

Table 3
Transition location on the center line of the sharp cone.

Cases	*AoA* (°)	Transition Onset Location (*TOL*)/mm			Deviation from experiments	
		experiments [27]	extended model	original model	extended model	original model
I	0	300	320	>800	6.7%	>166.7%
II	2	520	700	653	34.6%	–
III	4	600	663	591	10.1%	–
IV	6	620	679	357	9.5%	–

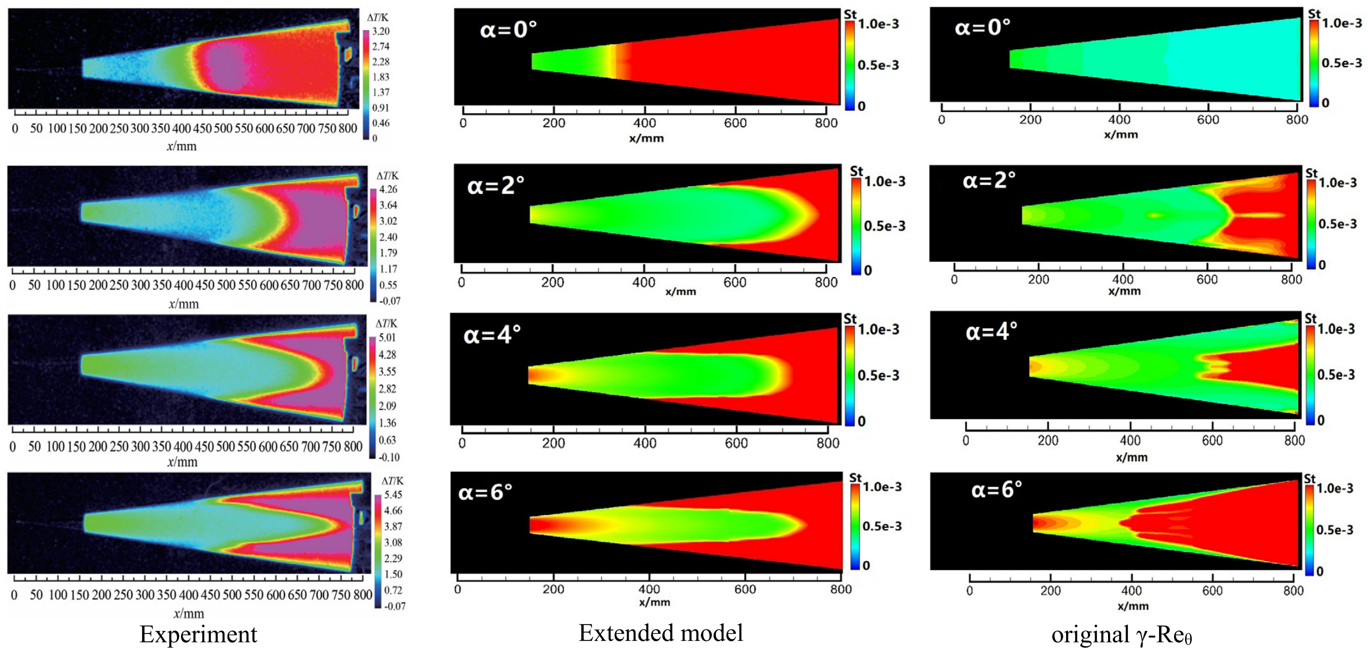

Fig. 7. Experimental temperature rise distribution [27] and model predicted St distribution.

Table 4
Freestream conditions of the elliptic cone [28].

Ma_∞	Re_∞ (/m)	*Noise Level*	Tu_∞ (for modeling)	h_{rsm} (μm)	α (∘)
6	10.3×10^6 11.3×10^6 13.0×10^6	≤0.05%	0.1%	0.3	0

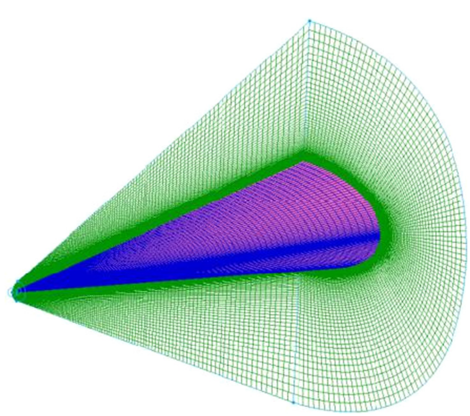

Fig. 8. Computed mesh for HIFiRE-5 elliptic cone.

based on the transition modes. The HyTRV's transition is investigated from three aspects: the cross-flow transition criterion, the transition on the windward side and the transition on the leeward side.

4.2. Computational details

The transition experiments of HyTRV have been carried in the ϕ2 m shock tunnel at CARDC (China Aerodynamic Research and Development Center). The HyTRV test model is half of the real size and the transition location is observed by a temperature sensitive paint (TSP) technique. In this work, the computation results are compared with stability analysis results [29] and part of the experimental results. The inflow conditions are summarized in Table 5. Among them, case I and case V have corresponding experimental data, all the cases have corresponding stability analysis results [29]. The inflow turbulence intensity is estimated from the noise level. The free stream turbulence intensity was assumed as $Tu_\infty = 0.8\%$ for transition modeling, the surface roughness height h = 1.6 μm. The computational mesh is half-model and contains about 3.2 million hexahedral cells, as shown in Fig. 11. The first grid size in the normal direction of the wall is strictly limited to ensure y^+ to be less than 1.0.

A grid convergence analysis of three varied meshes is conducted for HyTRV lifting body. The three meshes are marked as M1, M2 & M3. From M1 - M3, the grid points are about 1.1 million, 3.2 million and 5.7 million. The grid number of normal directions are 101,141,181, respectively. Fig. 12 compares the St distributions among the different grids; St represents the dimensionless Stanton number. The difference of heat flux exists in the transition and turbulence area. The St distribution of M1 differs a lot from the other two while the difference between M2 and M3 is acceptable. Thus, the medium mesh M2 is selected for the following numerical simulation.

4.3. Results and discussion

The cross-flow transition locations share a typical three-dimensional characteristic. 2-D line graphs of transition location are not comprehensive enough for 3-D cross-flow transitions. Hence, the numerical and experimental three dimensional graphs of transition locations are directly compared. Fig. 13 is the heat distribution of HyTRV surface predicted by the laminar solver, C-γ-Re_θ transition model and SST RANS model.

In this chapter, the cross-flow transition on the HyTRV surface is studied by utilizing C-γ-Re_θ transition model with comparison of stability analysis and wind tunnel experiment results. With flow structure analysis and cross-flow criterion analysis, the cause of cross-flow and the main cross-flow zones are pointed out; the cross-flow transition along variation of angle of attacks and Reynolds number is studied.

The experimental distribution of surface temperature rise is acquired by temperature-sensitive paint technique. This technique is developed by Hypervelocity Aerodynamics Institute of CARDC, and is cross referenced to other measurement techniques such as thin

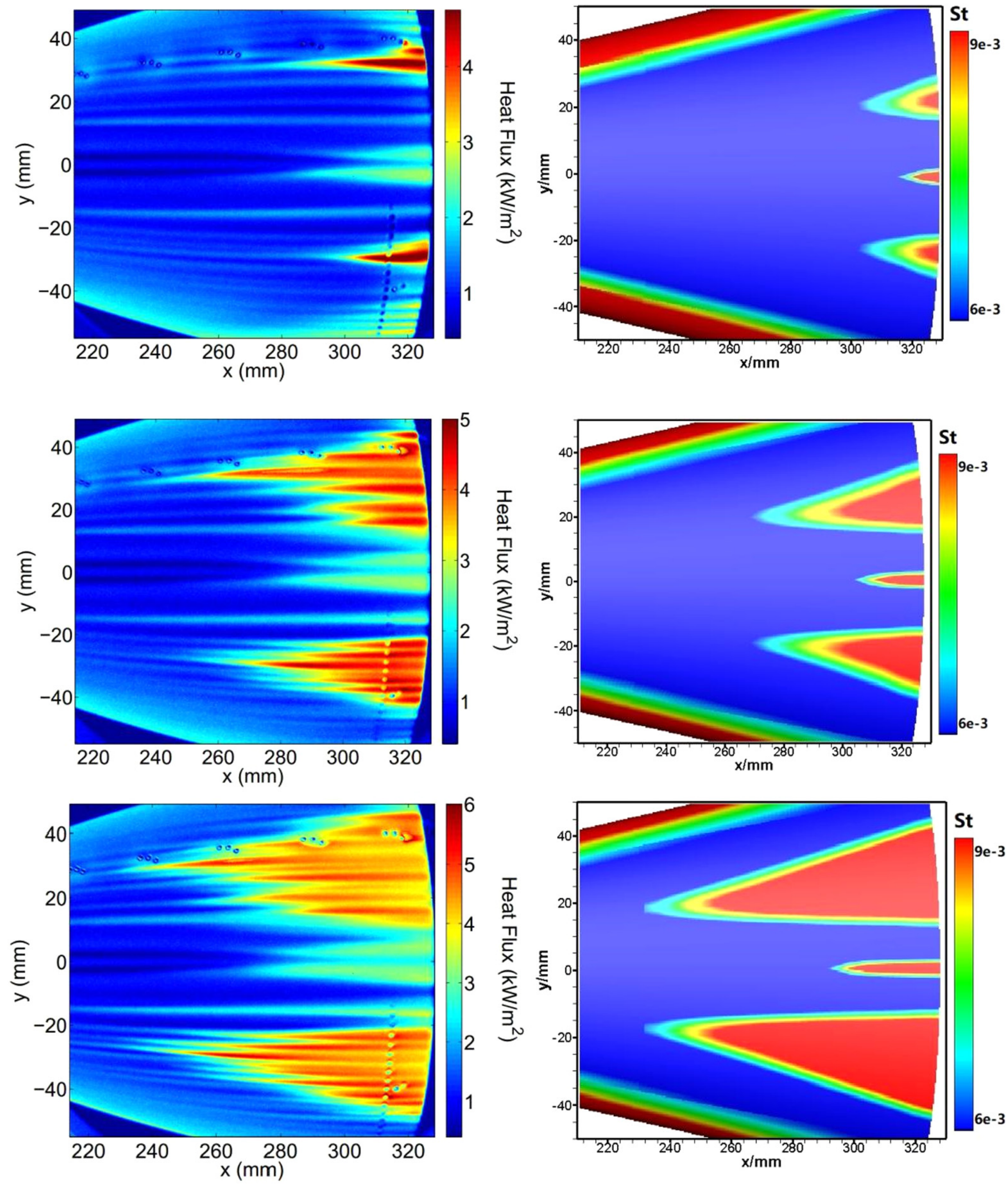

Fig. 9. Experimental heat-flux distribution [28] (left) and model predicted St distribution (right) on HIFiRE-5.

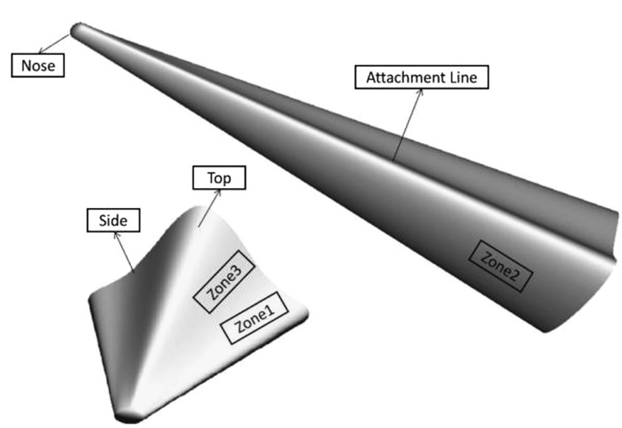

Fig. 10. Shape of HyTRV.

film heat flux sensor technique. The experiments of HyTRV are conducted in CARDC's ϕ2 m shock tunnel (FD-14A). The wind tunnel is composed of shock tube, nozzle, test section and vacuum chamber. The nozzle exist diameter is 1.2 m. The driving pressure reaches 50 MPa. The effective duration time is 4 ms to 18 ms with Ma 6-16 and Re 2.1×10^5-6.7×10^7/m. The HyTRV's experiments

Table 5
Test conditions and parameters of flow field.

Case	Ma_∞	Re_∞ (/m)	AoA (°)	Noise Level [30]	Tu_∞	P_0 (MPa)	T_0 (K)	T_∞ (K)	T_w (K)
I	6	1.1e7	0	3%	0.8%	2.67	795	97	300
II			2						
III			6						
IV			8						
V		4.2e7	0			10.4	664	81	

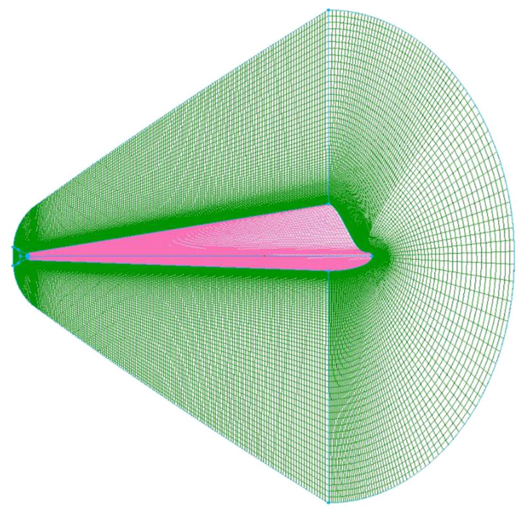

Fig. 11. Computed mesh for HyTRV lifting body.

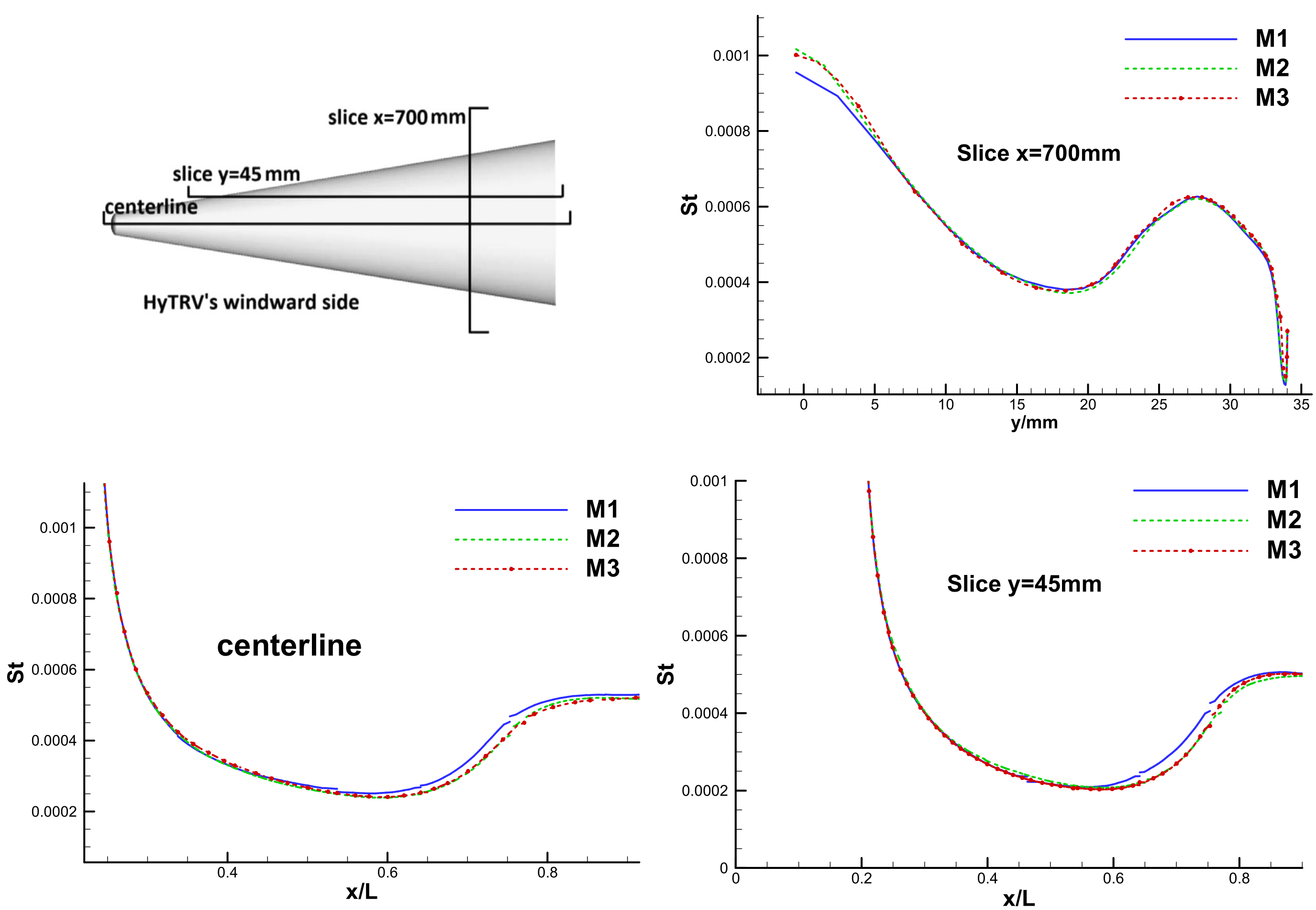

Fig. 12. St distributions of different grids for HyTRV lifting body.

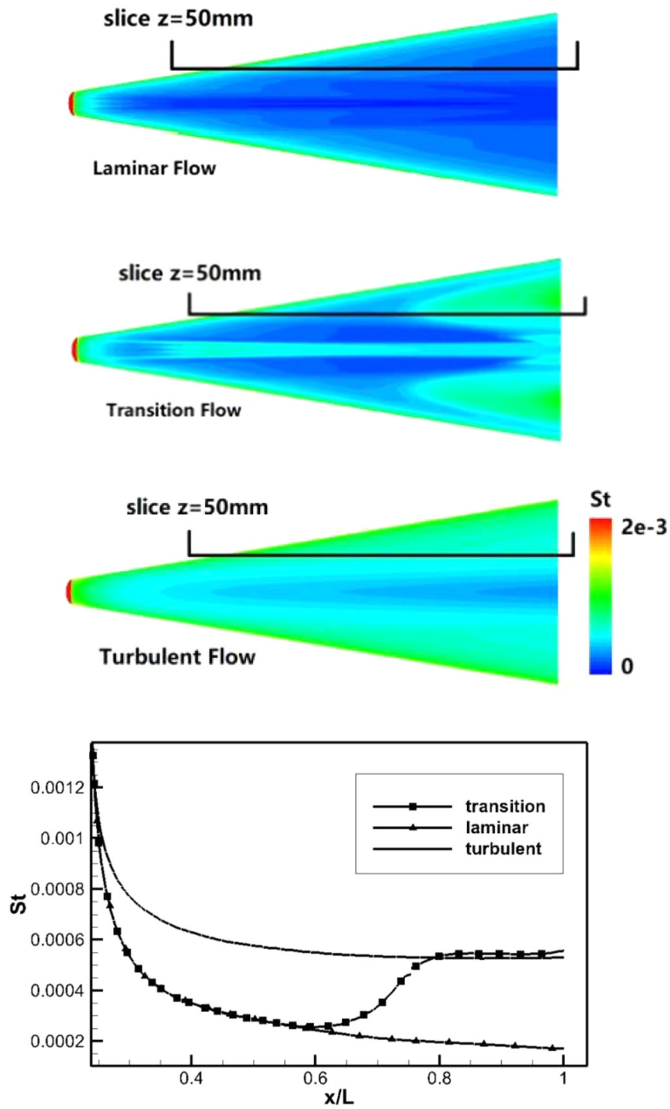

Fig. 13. *St* distribution of laminar, transition, and turbulent flows on HyTRV windward (Case I).

are mainly conducted at Ma=6 with wind tunnel's noise level to be about 3% [30].

The stability analysis with high precision simulations on HyTRV lifting body [29] is conducted by the "Transition and Complex Flow" research team of the State Key Laboratory of Aerodynamics. In their work, the eN method based on Linear Stability Theory (LST) is employed. Small disturbances can be assumed as traveling waves in the form of $q'(\xi,\eta,\zeta,t)=\widehat{q}(\eta)e^{i(\alpha\xi+\beta\zeta-\omega t)}+C.C$. In this equation, q' and $\widehat{q}$ denote the fluctuating physical quantities and the shape function. The physical quantity is $q=(\rho,u,v,w,T)^T$, where ρ, u, v, w, T denote density, streamwise velocity, azimuthal velocity and, temperature. For spatial disturbance wave, $\alpha=\alpha_r+i\alpha_i$, $\beta=\beta_r+i\beta_i$. The real parts α_r and β_r denote the dimensionless streamwise and azimuthal wave numbers. The imaginary parts α_i and β_i denote the dimensionless spatial growth rates. The ω denotes frequency. For a given frequency and wave number at one direction, the other can be obtained by iteration. For two dimensional flow, the propagation path at a specified frequency is determined along the inviscid streamline direction and the maximum growth rates can be obtained by integration of streamwise maximum spatial growth rate α_i along the path. For three-dimensional problems, the integration need to consider azimuthal wave numbers and their spatial changes. The integration method considering the equivalent spread wavenumber in the three dimensional boundary layer is adopted. The details of the method and solution can be found in [29].

4.3.1. Cross-flow transition criterion on HyTRV surface

The HyTRV's structure of shock waves, surface pressure distributions, and streamlines inside the boundary layer are illustrated in Fig. 14 and Fig. 15. The structure of shock wave in Fig. 14 contains the cross-section of the short and long axis, i.e., the side view and top view. At 0° angle of attack, shock waves at the top and side edge are closer to the surface than those at the centerline of the lower surface and concave areas of the upper surface. This leads the pressure on the former zones to be far higher than the later ones. In Fig. 15, the favorable pressure gradient direction zone is "side edge to centerline" on the lower surface. On the upper surface, the directions are "top to concave area" and "side edge to concave area". Thus, there are 2 circumferential favorable pressure gradient directions, one each on the upper and lower surfaces; this results in two pairs of cross-flow zones (Zone1 & Zone3) on the upper surface and one pair of the cross-flow zone (Zone2) on the lower surface. The solid line represents the near-wall stream-

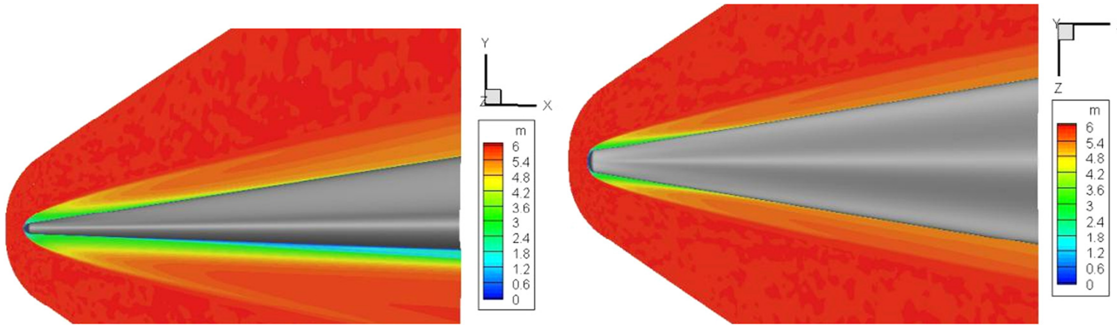

Fig. 14. Mach number contours of the cross sections of the short & long axes.

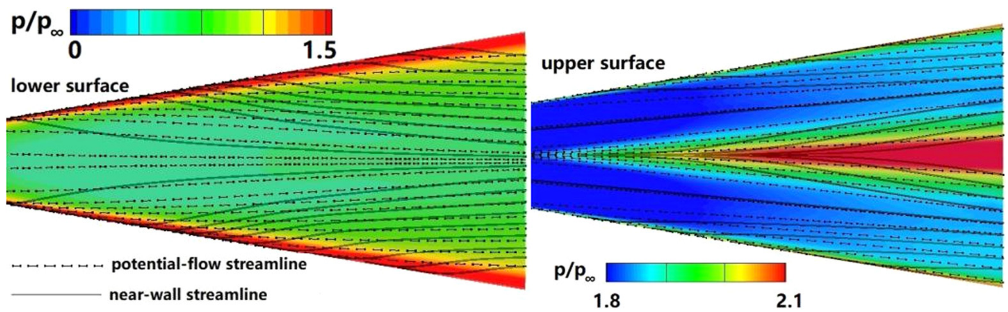

Fig. 15. Potential and near-wall streamlines of HyTRV in Case I (pressure contours on the surface).

line, the dashed line represents the potential-flow streamline. In the above areas, two types of streamlines have certain angle and cross-flow emerges.

The cross-flow zones can be distinguished by comparing near-wall and potential-flow streamlines. Further, it can be quantitatively identified by means of cross-flow Reynolds number distribution. The cross-flow Reynolds number is the most used criterion for cross-flow transition; however, it is not directly adopted in the current transition model for its unlocalized property. Further, it is one of the main methods to quantify cross-flow intensity. Owen and Randall [31] first proposed the transition criterion of cross-flow Reynolds number based on the experiment of the swept-wing. The definition of $\mathrm{Re}_{\mathrm{crossflow}}$ is given as:

$$\mathrm{Re}_{crossflow} = W_{\max}\delta_{10\%}/\nu$$

where $\mathrm{W}_{\max}$ denotes the maximum value of the cross-flow velocity searching from the direction normal to the wall. $\delta_{10\%}$ denotes the wall distance at the farther location where the cross-flow velocity is 10% of $\mathrm{W}_{\max}$. ν is the fluid kinematic viscosity. Fig. 16 shows the comparison of $Re_{crossflow}$ distribution and cross-flow N value distribution. Under hypersonic conditions, $Re_{crossflow}$ is still an empirical criterion. When compared with *e-N* method of stability theory, $Re_{crossflow}$ lacks direct physical connection with cross-flow transition. In Fig. 16, the two methods still show some consistency. The e-N predicted cross-flow zones on the upper (Zone1 & Zone3) and lower surface (Zone2) are reflected in $Re_{crossflow}$ distribution with a high value. The contour and position of high N value zones are consistent with those of $Re_{crossflow}$ distributions. In the figures of the upper surface, there are a pair of long sharp strips of high value areas both in N-value distributions and $Re_{crossflow}$ distributions. This can be attributed to the streamline convergence. The streamwise vortices significantly increase the boundary layer thickness. This causes the $\delta_{10\%}$ in the $Re_{crossflow}$ criterion to increase by even an order of magnitude, which leads to the emergence of strips area where the cross-flow intensity is not sufficient.

4.3.2. Transition predictions on the windward side

The previous analysis results indicate that both the e-N method and $Re_{crossflow}$ can be cross-flow transition criterion. Even though the e–N method is more physical, the two criteria cannot conduct a transition flow field simulation. The extended C-γ-Re_θ model is applied for HyTRV transition prediction. The model predicted heat distribution is analyzed with experimental results and e-N results.

The results at 0 degree *AoA* with two Reynolds numbers are analyzed in Fig. 17. The transition on the lower surface of HyTRV is dominated by cross-flow instabilities under a circumferential pressure gradient. The transition location is observed with temperature sensitive paint technique in the wind tunnel.

In Fig. 17a, the critical N value is calibrated to be 5.6 with inflow $Re = 1.1 \times 10^7$. The shape of the line with N-value to be 5.6 is basically the same as the experimental transition location. The frequency range of the N value is from 10-40 kHz, which indicates that the transition mode of the lower surface of the HyTRV in the current condition is dominated by cross-flow. Fig. 17c is the corresponding N-value contour of cross-flow with a frequency range from 0-30 kHz. The most unstable mode is at Zone2 with a similar location of experiments. In Fig. 17e, the predicted transition onset location is at x/Lt approximately equal to 0.75, which is consistent with Fig. 17a. L or Lt is the total length of HyTRV model. The overall shape of the model predicted *TOL* is slightly different from that of the experiments.

Figs. 17 (b, d, and f) illustrate the experimental measurements, stability analysis results, and model predicted results at $Re = 4.2 \times 10^7$. The low frequency signal of pressure pulsation in wind tunnel vanishes and flow condition tends to be more turbulent. Unlike the cases at $Re = 1.1 \times 10^7$ (Fig. 17 a c e) on the lower surface, the shape of the N-value contour differs considerably from the experimental results. It can be inferred that the modes on the lower surface of the experiment and stability analysis are different, and this requires further investigation. The model predicted transition location meets the wind tunnel well. The transition may not simply be whole-field cross-flow transition, which demonstrates

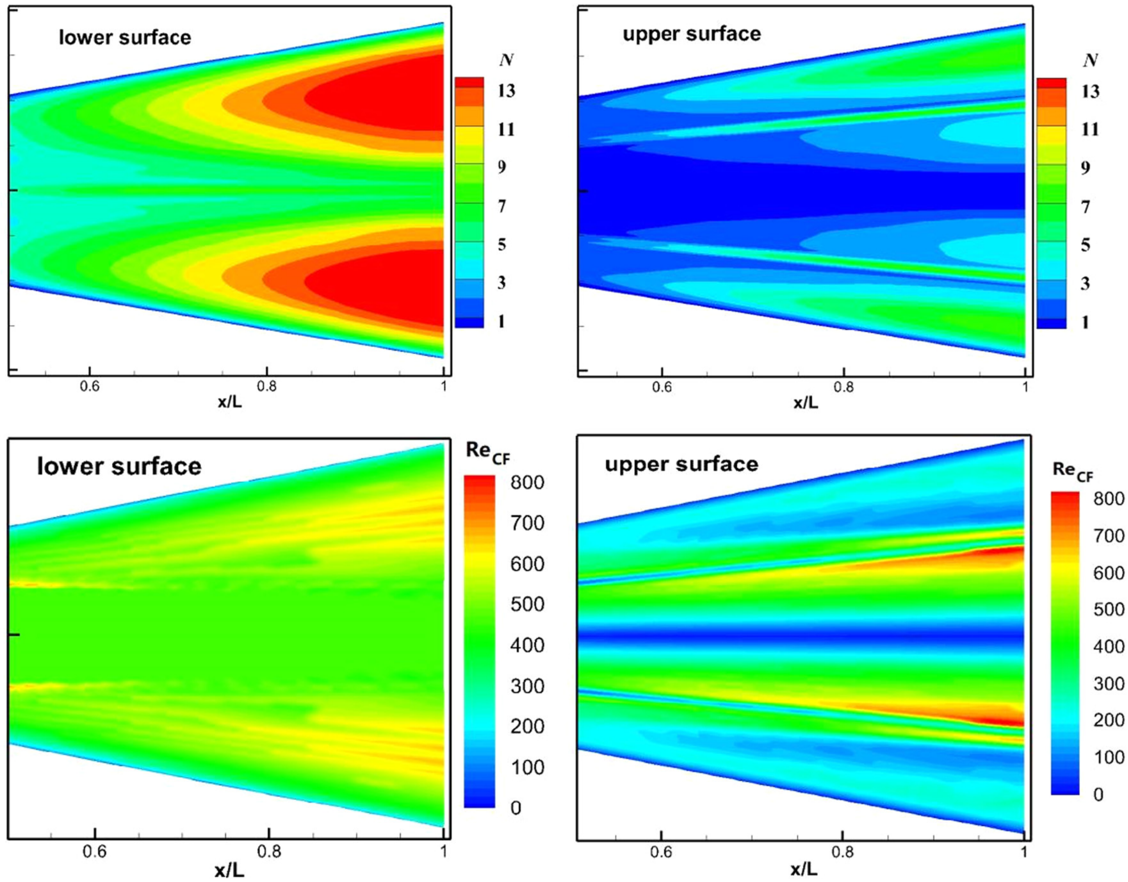

Fig. 16. Distribution of N value [29] and $Re_{crossflow}$ on HyTRV's surface (Case I).

the good potential of the C–γ–Re_θ model in the hypersonic 3D boundary layer transition prediction.

The C-γ-Re_θ model predicted *St* distribution and N-value distribution of stability theory are compared in Fig. 17, 18 & 19. The focus is on the high heat flux area (Zone2) induced by cross-flow transition. The long-strip-shape high heat flux zones on the top and side edges are caused by the near-wall shock wave, and they are not caused by the cross-flow transition.

From 0-8° *AoA* (Case I to IV), the N-value of stability analysis with a frequency range of 0-30 kHz is compared with transition prediction results in Fig. 18. The most unstable area on the lower surface is Zone2 [29]. The *N*-value decreases with the increase of *AoA*; the *N*-value contour retreats. The model predicted results are consistent with the trend of stability analysis results. Owing to the decrease in the circumferential pressure gradient caused by *AoA*; the cross-flow intensity decreases and *TOL* retreats until it finally disappears. In Cases 3 & 4 (Fig. 18 c, d), there is no model predicted cross-flow transition observed on Zone 2 and the *N*-value has not reached the critical number of N = 5.6. The cross-flow transition does not take place in the two cases above.

4.3.3. Transition predictions on the leeward side

Unlike the lower surface of HyTRV, the cross-section of the upper body is a curve controlled by CST (class function and shape function transformation technique) function and elliptic function. The common instability modes include cross-flow, Mack, and attachment line mode [29]. On the upper surface, the model prediction is compared to the stability analysis results to study the HyTRV's transition for the lack of experimental results on the upper surface.

The trend of *N*-value distribution on the upper surface with an increase in the *AoA* is different from the lower surface. When *AoA* increases from 0°, the N-value on the upper surface increases and the cross-flow area expands. The model predicted high heat flux area shares the same trend. However, Fig. 19 (c, d) show a model predicted long-strip-shape high heat flux area exists without the corresponding N-value contour. The transition model incorrectly turns on the D_{CF} because strong stream-wise vortices exist in that area. In this model, $H_{crossflow}$ is adopted to quantify cross-flow intensity. It is essentially a dimensionless stream-wise vorticity intensity and leads to inherent limitations of the model that the cross-flow term may be activated in the area of strong stream-wise vortices and incorrectly predicts the long-strip-shaped high heat flux zones.

5. Conclusions

The cross-flow extended C-γ-Re_θ transition model is developed and applied for the hypersonic three-dimensional boundary layer transition predictions on HyTRV lifting body. A preliminary investigation on model predictions with comparison of stability analyzed results and wind tunnel results is conducted. The flow characteristics and transition phenomenon of HyTRV lifting body are studied, and the transition model is validated in a typical aerodynamic configuration of a real flight vehicle. The conclusions are as follows:

(1) The hypersonic cross-flow extension is validated through hypersonic wind tunnel experiments of the sharp and the elliptic cones, and yield good predictions. In HyTRV lifting body's transition prediction, the transition criterion shares similar distribution with N-value of stability analysis. The model predicted *TOL*

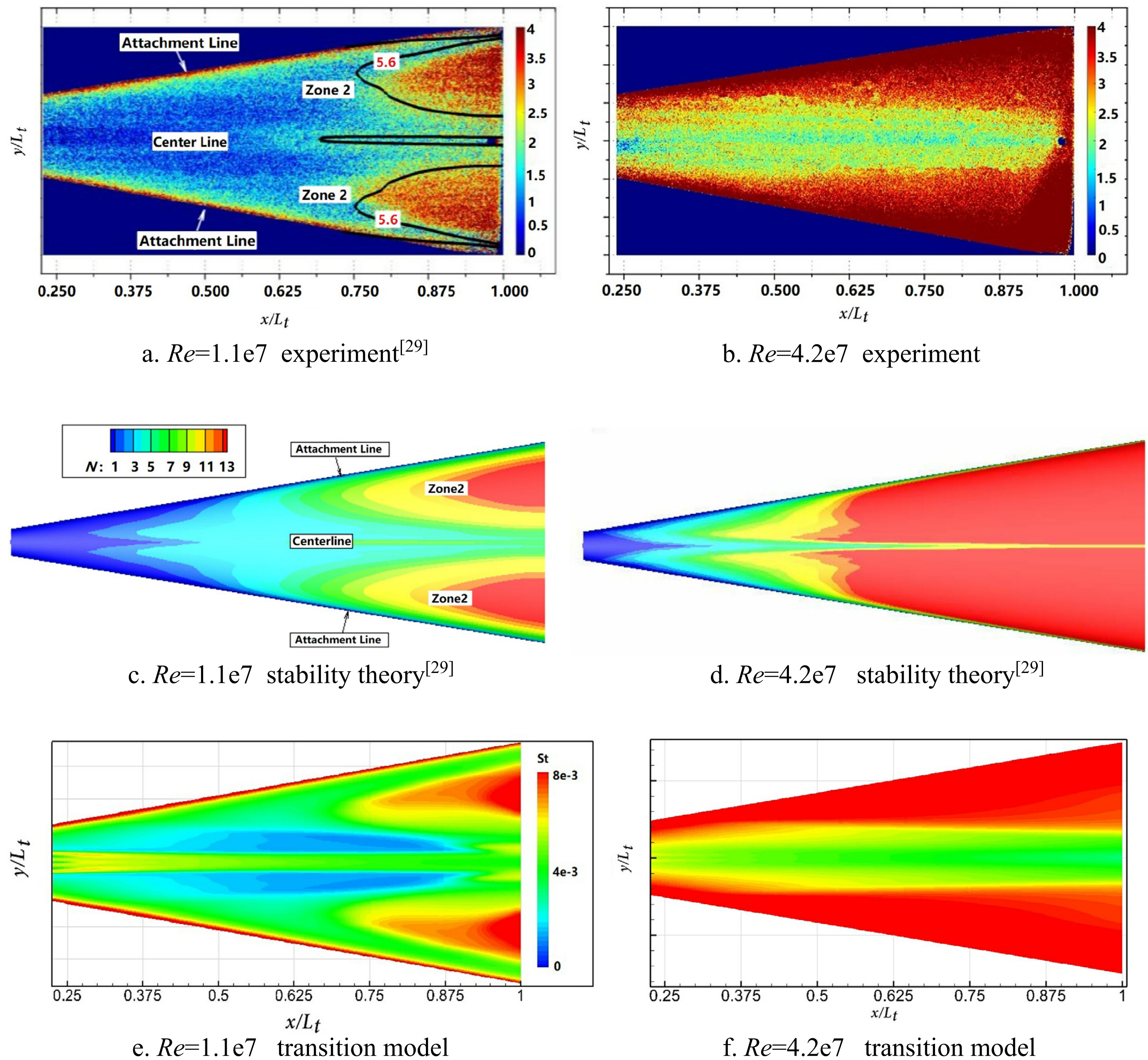

Fig. 17. Wind tunnel TSP results (a & b), N value distribution of stability theory (c & d) and heat flux distribution predicted by C-γ-Re_θ model (e & f) on HyTRV's lower surface (Case I & Case V).

matches the TSP wind tunnel results. The satisfactory validation results exhibit the extended model's potential for hypersonic cross-flow transition prediction.

(2) The comprehensive study of wind tunnel experiments, stability analysis and transition model indicates that a large area of cross-flow transition exists on the lower surface of HyTRV at *AoA* $= 0°$; further, several relatively small cross-flow transition zones exist on the upper surface. With the increase of *AoA*, circumferential pressure gradient of the lower surface decreases and *TOL* retreats, the gradient of upper surface increases and *TOL* moves forward. Several prominent cross-flow transition areas exist on HyTRV's surface, the HyTRV model can be a standard test model for hypersonic transition research.

(3) The limitations for C-γ-Re_θ transition model: Cross-flow intensity is approximated by non-dimensional cross-flow strength $H_{crossflow}$. On this condition, the area with a strong stream-wise vortex may be misjudged as a cross-flow area, the *TOL* moves forward and the long-strip irrational cross-flow transition zone may appear. The accurate definition and simulation of freestream disturbance still remain an open problem for transition modeling.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgements

The research work is supported by *National Key R&D Program* (Grant No. 2016YFA0401200) and grants from the *National Nature Science Foundation of China* (No.12002355, No.92052301, and No.11872370) and *National Numerical Windtunnel Project*.

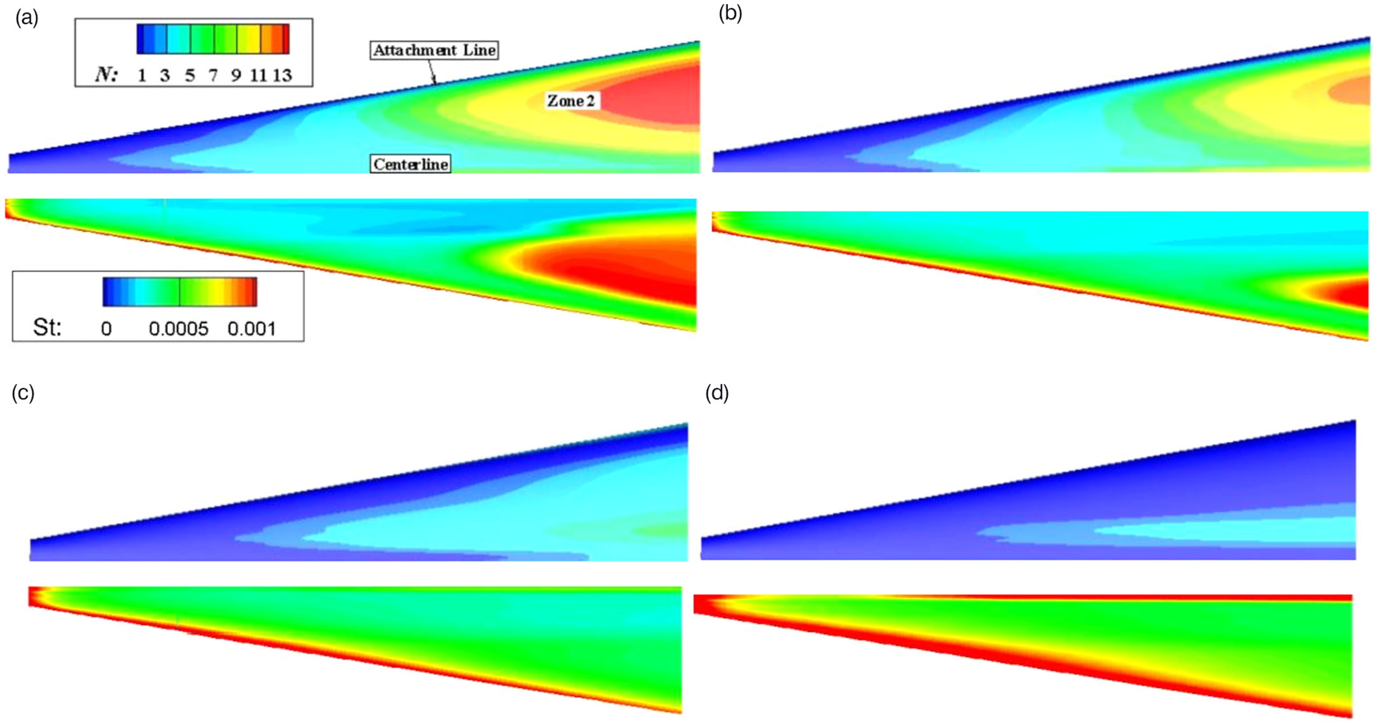

Fig. 18. N value distributions of cases I–IV windward side (upper) and C–γ–Re_θ heat flux distributions (lower).

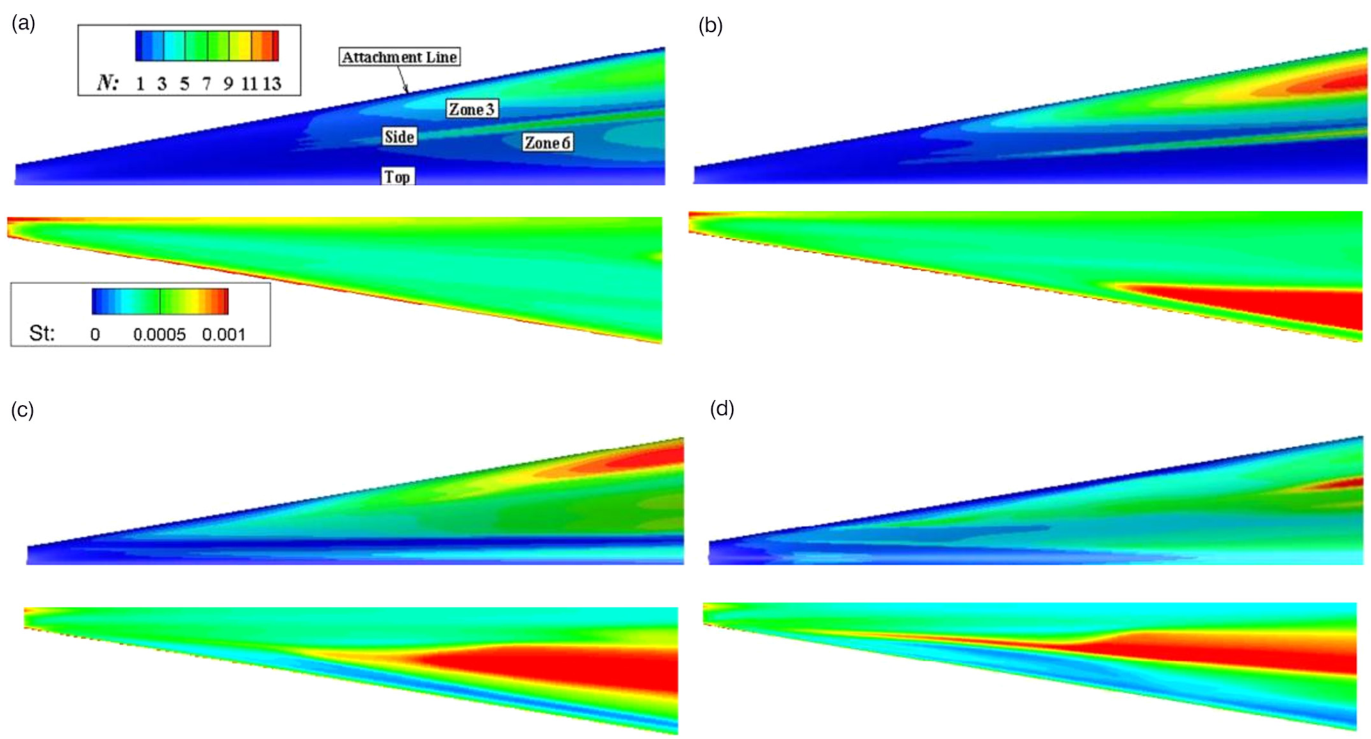

Fig. 19. N value distributions of Cases I–IV leeward side (upper) and C–γ–Re_θ heat flux distributions (lower).

References

[1] J.Q. Chen, G.H. Tu, Y.F. Zhang, G. Xu, X. Yuan, C. Chen, Hypersonic boundary layer transition: what we know, where shall we go, Acta Aerodyn. Sin. 35 (3) (2017) 311–337 (in Chinese).

[2] X.H. Xiang, Y.F. Zhang, J.Q. Chen, et al., Progress in transition models for cross-flow instabilities, Acta Aerodyn. Sin. 36 (2) (2018) 254–264 (in Chinese).

[3] L. Zhou, Z. Gao, Y. Du, Flow-dependent DDES/γ-Reθ coupling model for the simulation of separated transitional flow, Aerosp. Sci. Technol. 87 (2019) 389–403, https://doi.org/10.1016/j.ast.2019.02.037.

[4] J. Han, J.Y. Hwang, O.J. Kwon, Numerical investigation of effect of crossflow transition on rotor blade performance in hover, Aerosp. Sci. Technol. (2021) 106733 (online).

[5] L. Chen, Z. Guo, X. Deng, Z. Hou, Aerodynamic performance and transition prediction of low-speed fixed-wing unmanned aerial vehicles in full configuration based on improved γ-Reθ model, Aerosp. Sci. Technol. 107 (2020) 106281.

[6] X.H. Xiang, J.Q. Chen, X.X. Yuan, et al., Study on C-γ-Re_θ transition model for hypersonic cross-flow transition prediction, Acta Aeronaut. Astronaut. Sin. 42 (9) (2021) 625711, https://doi.org/10.7527/S1000-6893.2021.25711 (in Chinese).

[7] X. Zhang, Z. Gao, B. Zhong, Extensions of Menter's SST transition model to simulate hypersonic and cross-flow transition, in: Int. Cong. Aeronaut. Sci., 2012.

[8] X. Zhang, Z. Gao, A numerical research on a compressibility-correlated Langtry's transition model for double wedge boundary layer flows, Chin. J. Aeronaut. 24 (3) (2011) 249–257, https://doi.org/10.1016/s1000-9361(11)60030-7.

[9] L. Wang, S. Fu, Development of an intermittency equation for the modeling of the supersonic/hypersonic boundary layer flow transition, Flow Turbul. Combust. 87 (1) (2011) 165–187, https://doi.org/10.1007/s10494-011-9336-1.

[10] L. Zhou, C. Yan, Z.H. Hao, W.X. Kong, Y. Zhou, Transition model and transition criteria for hypersonic boundary layer flow, Acta Aeronaut. Astronaut. Sin. 37 (2016) 1092–1102.

[11] Y.T. Zhao, H.K. Liu, Q. Zhou, C. Yan, Quantification of parametric uncertainty in k-ω-γ transition model for hypersonic flow heat transfer, Aerosp. Sci. Technol. 96 (2020) 105553, https://doi.org/10.1016/j.ast.2019.105553.

[12] Z. Liu, Y. Zhao, S. Chen, C. Yan, F. Cai, Predicting distributed roughness induced transition with a four-equation laminar kinetic energy transition model, Aerosp. Sci. Technol. 99 (2020) 105736.

[13] Y. Zhang, Y. Zhang, J. Chen, M. Mao, Y. Jiang, Numerical simulations of hypersonic boundary layer transition based on the flow solver chant 2.0, in: 21st AIAA Int. Space Planes and Hypersonics Technol. Conf., 2017, p. 2409.

[14] J.L. Xu, Y.F. Song, Y. Zhang, et al., Turbulence mixing length for compressible free shear flows, Chin. J. Aeronaut. 37 (6) (2016) 1841–1850.

[15] J.L. Xu, D. Chen, Y. Song, S. Ji, Y. Zhou, Mean flow compressibility effects in transonic turbulence modeling, Aerosp. Sci. Technol. 79 (2018) 492–499, https://doi.org/10.1016/j.ast.2018.06.005.

[16] J. Xu, J. Bai, L. Qiao, Y. Zhang, Development of a computational fluid dynamics compatible mathematical model for boundary layer transitional flows in low-disturbance environment, Aerosp. Sci. Technol. 86 (2019) 487–496, https://doi.org/10.1016/j.ast.2019.01.036.

[17] R.B. Langtry, K. Sengupta, D.T. Yeh, et al., Extending the Local Correlation Based Transition Model for Crossflow Effects, AIAA Paper, 2015, AIAA-2015-2474.

[18] X.H. Xiang, H.J. Ren, Y.F. Zhang, et al., Transition prediction with hypersonic cross-flow model on HIFiRE-5, J. Phys. Conf. Ser. 1786 (2021) 012051.

[19] H. Bippes, B. Müller, M. Wagner, Measurements and stability calculations of the disturbance growth in an unstable three-dimensional boundary layer, Phys. Fluids A, Fluid Dyn. 3 (10) (1991) 2371–2377, https://doi.org/10.1063/1.858176.

[20] T. Juliano, S. Schneider, Instability and transition on the HIFiRE-5 in a Mach 6 quiet tunnel, in: 40th Fluid Dyn. Conf. Exhibit, 2010, p. 5004.

[21] Y.F. Zhang, J. Lei, Y.R. Zhang, M.L. Mao, J.Q. Chen, Calibration of transition model for hypersonic numerical simulation platform, Acta Aerodyn. Sin. 33 (1) (2015) 42–47 (in Chinese).

[22] C. Müller, F. Herbst, Modelling of crossflow-induced transition based on local variables, in: Proc. ECCOMAS, Vol. 2252, 2014, p. 72.

[23] D.J. Dinzl, G.V. Candler, Direct simulation of hypersonic crossflow instability on an elliptic cone, AIAA J. 55 (6) (2017) 1769–1782, https://doi.org/10.2514/1.j055130.

[24] T.J. Juliano, R.L. Kimmel, S. Willems, A. Guelhan, R. Wagnild, Hifire-1 boundary-layer transition: ground test results and stability analysis, in: 53rd AIAA Aerosp. Sci. Meet., 2015, p. 1736.

[25] X.H. Xiang, Y.F. Zhang, J.Q. Chen, et al., Uncertainty quantification analysis and application research on cross-flow transition parameters, J. Astronaut. 41 (9) (2020) 1141–1150.

[26] S.S. Chen, F.J. Cai, H.C. Xue, N. Wang, C. Yan, An improved AUSM-family scheme with robustness and accuracy for all Mach number flows, Appl. Math. Model. 77 (2020) 1065–1081, https://doi.org/10.1016/j.apm.2019.09.005.

[27] J.F. Chen, G. Ling, Q.H. Zhang, et al., Infrared thermography experiments of hypersonic boundary-layer transition on a 7° half-angle sharp cone, J. Exp. Fluid Mech. 34 (1) (2020) 60–66.

[28] M.P. Borg, R.L. Kimmel, Simultaneous infrared and pressure measurements of crossflow instability modes for HIFiRE-5, in: 54th AIAA Aerosp. Sci. Meet., 2016, p. 0354.

[29] J.Q. Chen, G.H. Tu, B.B. Wan, Characteristics of flow field and boundary-layer stability of Hypersonic Transition Research Vehicle (HyTRV), J. Aeronaut. 42 (4) (2021) 124317 (in Chinese).

[30] Q. Li, T. Jiang, S.Y. Chen, et al., Measurement technique and application of boundary layer transition in shock tunnel, Acta Aeronaut. Astronaut. Sin. 40 (8) (2019) 122740, https://doi.org/10.7527/S1000-6893.2019.22740 (in Chinese).

[31] P.R. Owen, D.G. Randall, Boundary layer transition on a sweptback wing, Rep. No. Aero 277, British RAE, May 1952.

[32] X.J. Yuan, X.G. Sha, X.T. Shi, et al., Noise-turbulence relationship in hypersonic flow, Acta Aeronaut. Astronaut. Sin. 41 (11) (2020) 123791, https://doi.org/10.7527/S1000-6893.2020.23791 (in Chinese).